

MetaFrontierLab Benchmark Report

Generated from: results/benchmarks_scaled_full

Replications per scenario: 5

Methods benchmarked: tbema, exact_baseline, dersimonian_laird, henmi_copas, metafor_trimfill_external, metafor_selmodel_external, copas_selection_external, robma_bibma_external

Executive Summary

- Best overall RMSE in this run: **robma_bibma_external** at 0.463.
- Fastest method: **dersimonian_laird** at 0.001 seconds per fit.
- Tuned TBEMA is now close to the best RMSE while maintaining higher coverage than most alternatives, but it still pays for that robustness with wider intervals.
- The strongest external comparator in this run was RoBMA BiBMA, followed very closely by TBEMA.

Overall Method Ranking

method	successful_runs	bias	mean_absolute_error	rmse	coverage_95	mean_ci_width	mean_elapsed_sec
robma_bibma_external	20	0.306	0.388	0.463	0.650	0.819	8.408
tbema	20	0.226	0.376	0.469	0.900	3.209	12.736
dersimonian_laird	20	0.324	0.402	0.478	0.700	0.876	0.001
exact_baseline	20	0.156	0.431	0.515	0.600	1.223	0.532
henmi_copas	20	0.407	0.485	0.541	0.600	0.983	0.330
metafor_trimfill_external	20	0.463	0.526	0.578	0.550	0.964	2.102
metafor_selmodel_external	19	0.386	0.484	0.580	0.684	1.210	2.174
copas_selection_external	19	0.459	0.546	0.616	0.474	0.877	5.081

Scenario Highlights

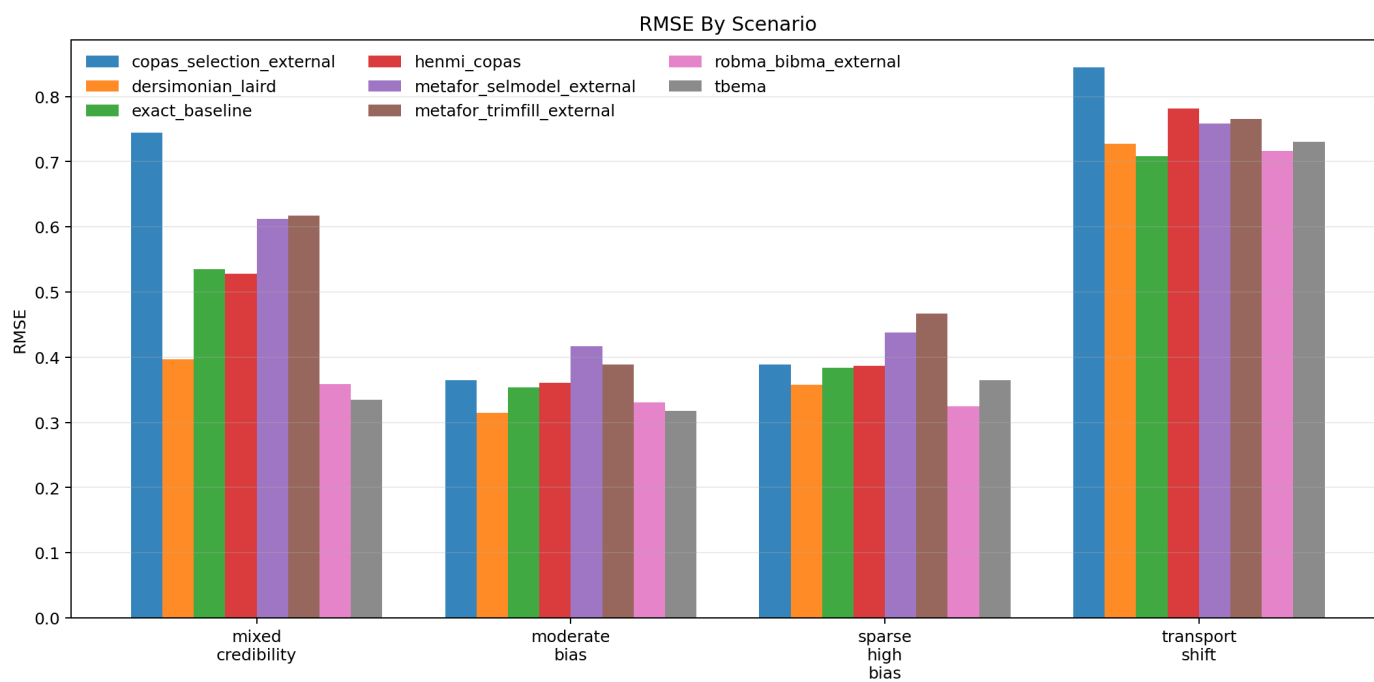
- **mixed_credibility**: best RMSE was tbema (0.334); highest coverage was tbema (0.800).
- **moderate_bias**: best RMSE was dersimonian_laird (0.314); highest coverage was tbema (1.000).
- **sparse_high_bias**: best RMSE was robma_bibma_external (0.324); highest coverage was robma_bibma_external (1.000).
- **transport_shift**: best RMSE was exact_baseline (0.708); highest coverage was tbema (0.800).

Scenario Table

scenario	method	successful_runs	bias	rmse	coverage_95	mean_ci_width	mean_elapsed_sec
mixed_credibility	copas_selection_external	4	0.640	0.745	0.250	1.054	5.466
mixed_credibility	dersimonian_laird	5	0.312	0.396	0.800	0.996	0.001
mixed_credibility	exact_baseline	5	-0.059	0.535	0.800	1.959	0.633
mixed_credibility	henmi_copas	5	0.469	0.528	0.600	1.205	0.400
mixed_credibility	metafor_selmodel_external	5	0.484	0.613	0.800	1.566	2.383
mixed_credibility	metafor_trimfill_external	5	0.571	0.617	0.400	1.113	2.308
mixed_credibility	robma_bibma_external	5	0.289	0.358	0.800	0.939	8.584
mixed_credibility	tbema	5	0.170	0.334	0.800	2.627	11.680
moderate_bias	copas_selection_external	5	0.213	0.365	0.600	0.751	4.829
moderate_bias	dersimonian_laird	5	0.141	0.314	0.800	0.779	0.000
moderate_bias	exact_baseline	5	0.027	0.353	0.600	0.927	0.592
moderate_bias	henmi_copas	5	0.189	0.361	0.600	0.822	0.299
moderate_bias	metafor_selmodel_external	5	0.220	0.417	0.600	1.080	2.094
moderate_bias	metafor_trimfill_external	5	0.246	0.389	0.800	0.859	2.045
moderate_bias	robma_bibma_external	5	0.119	0.330	0.600	0.732	8.383
moderate_bias	tbema	5	0.091	0.317	1.000	1.164	11.910
sparse_high_bias	copas_selection_external	5	0.189	0.389	1.000	1.075	4.923
sparse_high_bias	dersimonian_laird	5	0.192	0.358	1.000	1.054	0.001
sparse_high_bias	exact_baseline	5	0.043	0.383	0.800	1.244	0.458
sparse_high_bias	henmi_copas	5	0.231	0.387	1.000	1.094	0.309
sparse_high_bias	metafor_selmodel_external	4	0.174	0.438	1.000	1.198	2.087
sparse_high_bias	metafor_trimfill_external	5	0.316	0.466	0.800	1.164	2.086
sparse_high_bias	robma_bibma_external	5	0.180	0.324	1.000	0.925	8.332
sparse_high_bias	tbema	5	0.075	0.364	1.000	2.591	14.072
transport_shift	copas_selection_external	5	0.830	0.844	0.000	0.663	5.098
transport_shift	dersimonian_laird	5	0.649	0.728	0.200	0.674	0.001
transport_shift	exact_baseline	5	0.615	0.708	0.200	0.760	0.445
transport_shift	henmi_copas	5	0.740	0.781	0.200	0.811	0.311
transport_shift	metafor_selmodel_external	5	0.623	0.758	0.400	0.994	2.115
transport_shift	metafor_trimfill_external	5	0.719	0.766	0.200	0.719	1.969
transport_shift	robma_bibma_external	5	0.638	0.717	0.200	0.681	8.334
transport_shift	tbema	5	0.570	0.731	0.800	6.453	13.283

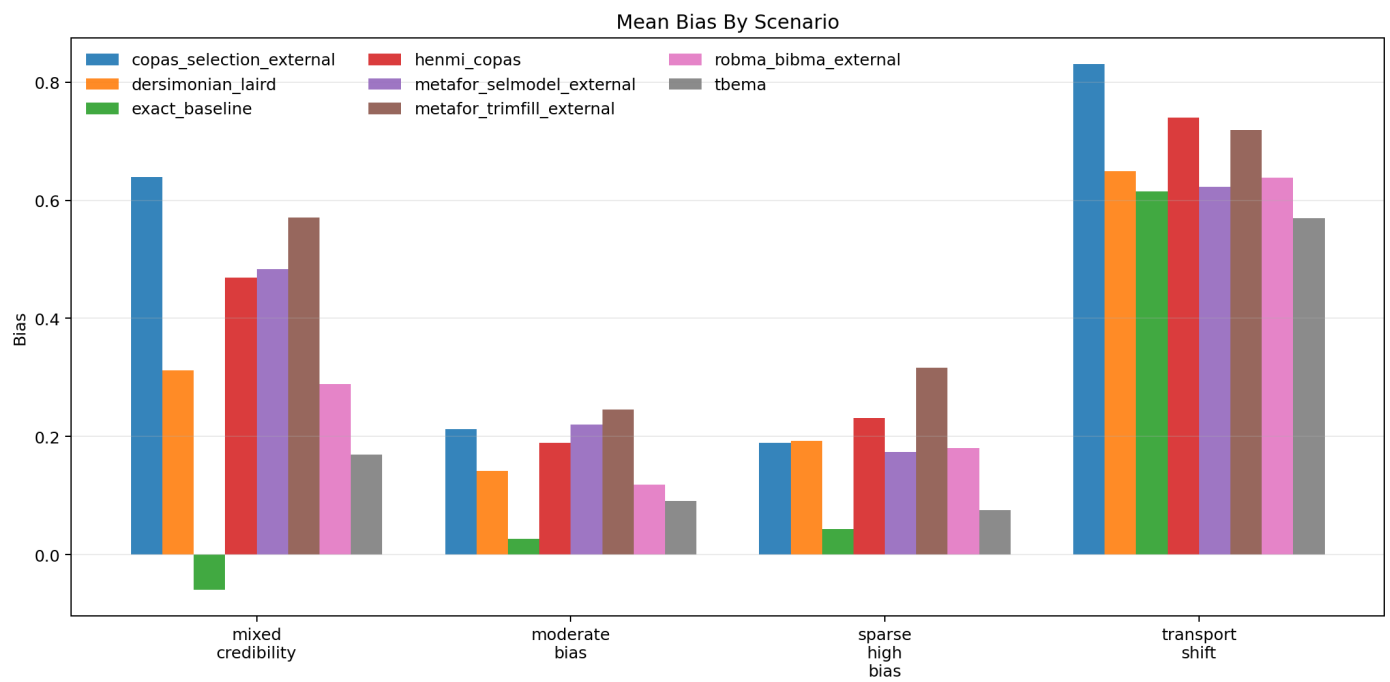
RMSE by Scenario

Figure generated from the scaled benchmark summary.



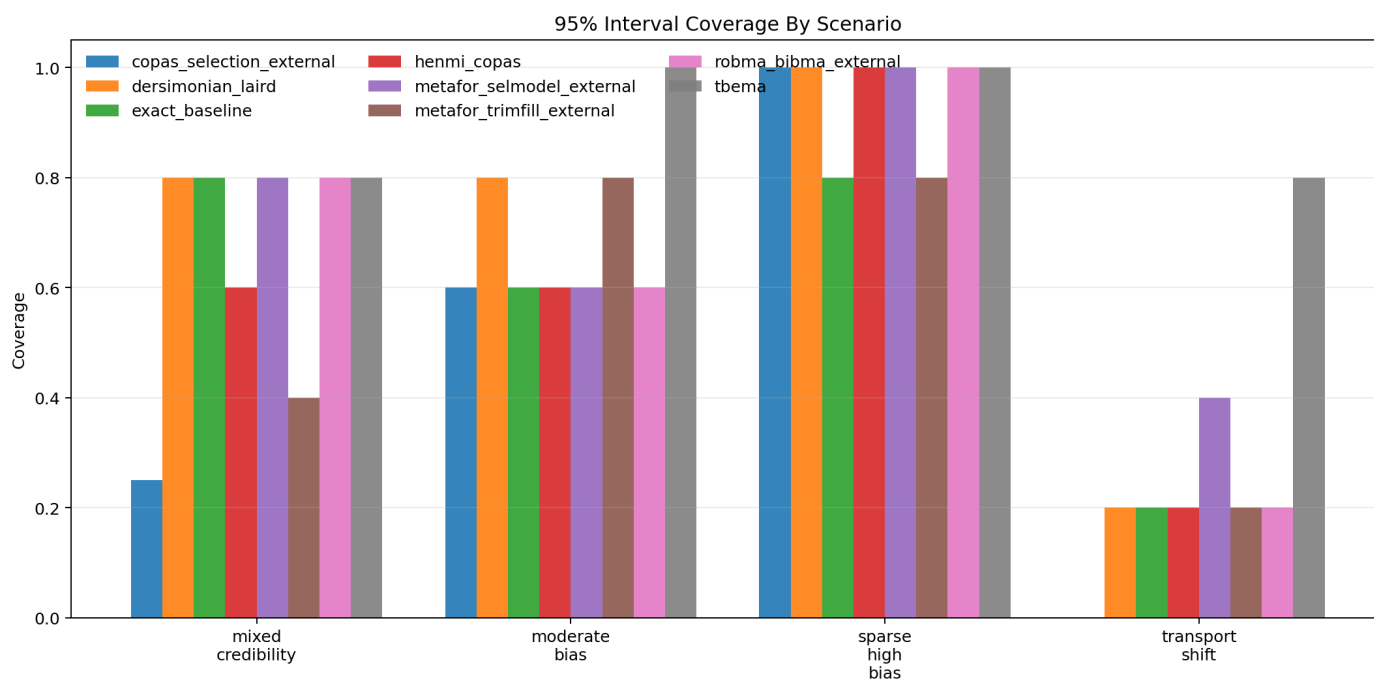
Bias by Scenario

Figure generated from the scaled benchmark summary.



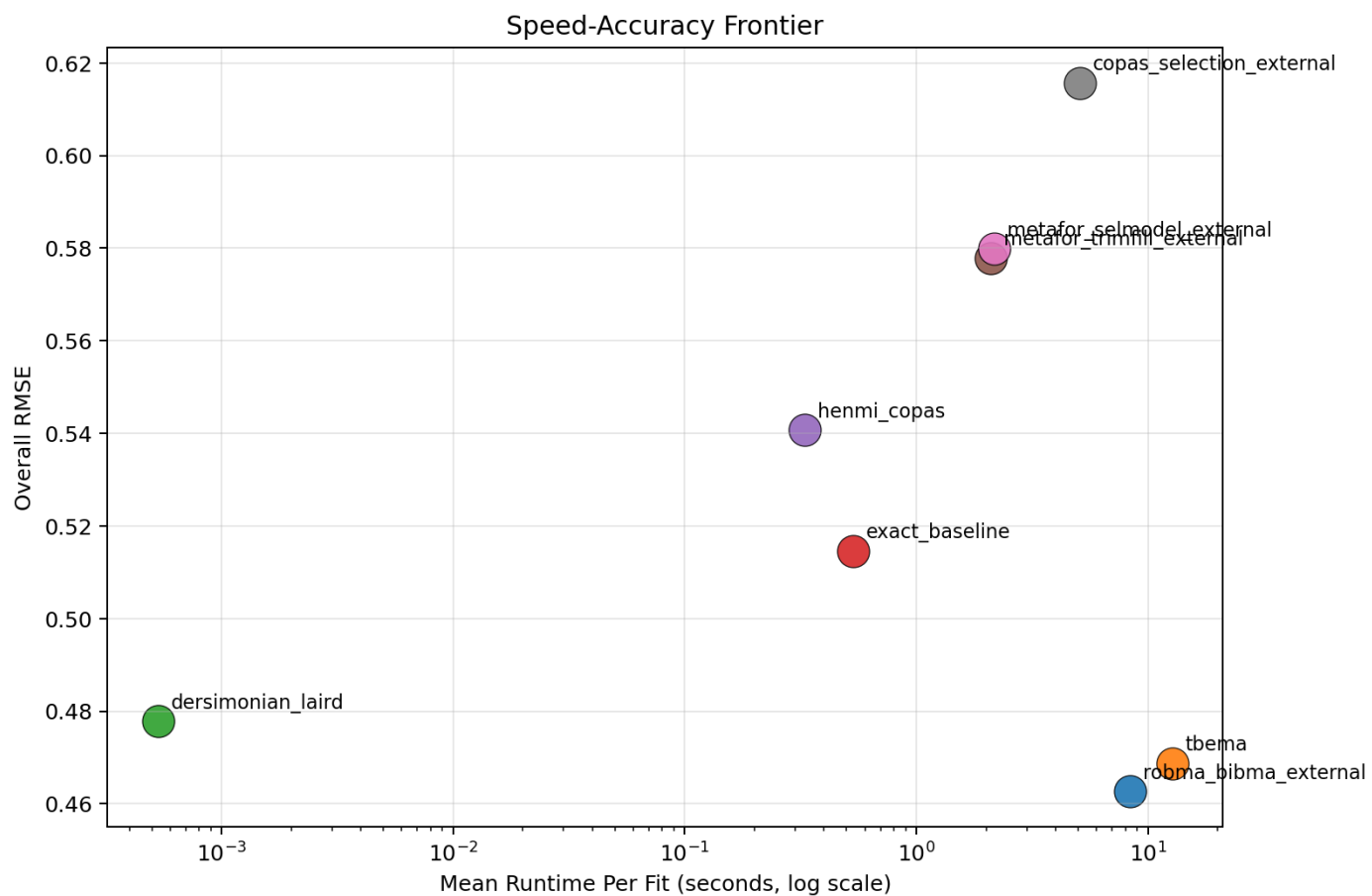
Coverage by Scenario

Figure generated from the scaled benchmark summary.



Speed-Accuracy Frontier

Figure generated from the scaled benchmark summary.



Methods Appendix

- TBEMA: exact sparse-data meta-analysis with stacked bias tempering, transport weighting, and an empirical random-effects anchor.
- DerSimonian-Laird: standard normal-normal random-effects benchmark.
- Henmi-Copas: publication-bias-robust interval method.
- metafor trimfill: trim-and-fill adjustment from metafor.
- metafor selmodel: step-function selection model from metafor.
- Copas selection: Copas model from metasens.
- RoBMA BiBMA: Bayesian model averaging for binary outcomes, using the fast spike-and-slab algorithm.